

DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: Natural Language Processing | Course Code: 21AI62 | Credits: 3

Exam Session: End Semester Exam 2024

Max Marks: 80 | Duration: 3 Hours

INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

- Q1.** Explain the Transformer Self-Attention mechanism mathematically:
 $\text{Attention}(Q, K, V) = \text{softmax}((Q K^T) / \sqrt{d_k}) * V$ and why scaling factor $\sqrt{d_k}$ is required.

[8 Marks]